ITA0609-MACHINE LEARNING

ASSESSMENT TOOL 1 – Application-Based Questions (3 Questions)

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1. A parking facility has three sections containing cars of different colors. Section A has

5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section

C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is

twice as likely to be selected as each of the other sections due to higher usage. A

randomly selected car from the chosen section turns out to be blue. What is the

probability that the car came from Section C?

Given:

Section	Red	Blue	Black	Total Cars
A	5	3	2	10
B	4	6	5	15
C	7	2	6	15

Since Section B is twice as likely:

$$P(A) = \frac{1}{4}, P(B) = \frac{1}{2}, P(C) = \frac{1}{4}$$

Probability of selecting a blue car:

$$P(Blue|A) = \frac{3}{10} P(Blue|B) = \frac{6}{15} = \frac{2}{5} P(Blue|C) = \frac{2}{15}$$

Using Bayes' Theorem:

$$P(C|Blue) = \frac{P(Blue|C)P(C)}{P(Blue)}$$

First find $P(Blue)$:

$$P(Blue) = \frac{3}{10} \times \frac{1}{4} + \frac{2}{5} \times \frac{1}{2} + \frac{2}{15} \times \frac{1}{4} = \frac{3}{40} + \frac{1}{5} + \frac{1}{30} = \frac{9+24+4}{120} = \frac{37}{120}$$

Now:

$$P(C|Blue) = \frac{\frac{2}{15}\times\frac{1}{4}}{\frac{37}{120}} = \frac{1}{30} \times \frac{120}{37} = \frac{4}{37}$$

Answer:

$$P(C|Blue) = \frac{4}{37} \approx 0.1081$$

Probability = 10.81%

2. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs: (5 Marks)

1. (a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4
2. (b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2
3. (c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
4. (d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
5. (e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
6. (f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
7. (g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
8. (h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2
9. i) Identify the epoch at which the model begins to overfit.
10. ii) Suggest two effective techniques to reduce overfitting in this model.
11. iii) Discuss how overfitting affects the model’s generalization ability on unseen data.

SOL.

2. Training Loss & Validation Loss Analysis (5 Marks)

Epoch	Training Loss	Validation Loss
1	2.8	2.4
2	2.5	2.2
3	2.2	2.0
4	2.0	2.1
5	1.8	2.3
6	1.6	2.6
7	1.5	2.9
8	1.3	3.2

i) Epoch at which overfitting begins

Validation loss decreases until Epoch 3 and starts increasing from Epoch 4.

$$Overfitting\ begins\ at\ Epoch\ 4$$

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ii) Two techniques to reduce overfitting

1. **Early Stopping**
   - Stop training when validation loss starts increasing.
2. **Regularization / Dropout**
   - Prevents the model from memorizing training data.

$$Techniques:\ Early\ Stopping\ and\ Dropout\ (Regularization)$$

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iii) Effect of overfitting on generalization

- Model learns noise in training data.
- Performs well on training data.
- Performs poorly on unseen data.
- Validation loss increases while training loss decreases.

$$Overfitting\ reduces\ the\ model's\ ability\ to\ generalize\ to\ new\ data.$$

3.Examine the performance of a machine-learning model using training and validation

accuracy over multiple epochs: (5 Marks)

12. (a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%

13. (b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%

14. (c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%

15. (d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%

16. (e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%

17. (f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%

18. (g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%

19. (h)Epoch 8: Training Accuracy = 92%,Validation Accuracy = 68%

20. i) Identify the epoch at which the model starts overfitting.

21. ii) Explain why validation accuracy decreases despite training

accuracy increasing. 22. iii) Suggest two methods to improve validation

performance.

Training Accuracy & Validation Accuracy Analysis (5 Marks)

SOL.

Epoch	**Training Accuracy**	**Validation Accuracy**
1	60%	58%
2	65%	63%
3	70%	68%
4	75%	72%
5	80%	73%
6	85%	72%
7	88%	70%

Epoch	Training Accuracy	Validation Accuracy
8	92%	68%

i) Epoch at which overfitting starts

Validation accuracy reaches maximum at Epoch 5 (73%) and decreases afterward.

$$Overfitting\ starts\ at\ Epoch\ 6$$

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ii) Why does validation accuracy decrease while training accuracy increases?

- The model starts memorizing training data.
- It captures noise rather than useful patterns.
- Hence training accuracy increases but performance on unseen data decreases.

$$The\ model\ memorizes\ training\ data\ and\ loses\ generalization\ ability.$$

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iii) Two methods to improve validation performance

1. **Data Augmentation / More Training Data**
2. **Early Stopping**

Other acceptable answers:

- Dropout
- L1/L2 Regularization
- Cross Validation

$$Methods:\ Early\ Stopping\ and\ Data\ Augmentation$$